ModelIt!

Student Activity Pack

Why Don't We See Dinosaurs?

What Fossils Tell Us About Long Ago

Grade 2 | NGSS 2-LS4-1 | Stability and Change

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you know about dinosaurs? Did they really exist?

2 What is a fossil? Have you ever seen one in a museum or a picture?

3 Draw what you think happened to the dinosaurs — why are they not here anymore?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Habitat Change | Animal Survival | Population Size

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Small Habitat Change

Habitat Change set to small

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Huge Habitat Change

Habitat Change set to very large

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Medium Change

Habitat Change set to medium

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Food Supply — How much food is available in the habitat — when food runs out, animals struggle to survive

Climate Speed — How fast the habitat changes — slow changes let animals adapt, but fast changes are harder to survive

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If dinosaurs lived on Earth millions of years ago, why are there no dinosaurs alive today — and how do we know they existed? Use evidence from your simulation to support your answers.

1 How do fossils help us learn about animals we have never seen alive?

2 If habitats are still changing today, could any animals alive now become extinct in the future?

8 STEM Challenge: Create a Fossil Museum Exhibit

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YOUR ENGINEERING MISSION

Design a mini museum exhibit that uses fossil evidence to teach visitors about an extinct animal.

THE SCENARIO

A children's museum wants to create a new exhibit about extinct animals. Your team has been hired to choose one extinct animal, make a model fossil, and create a display that teaches visitors what the animal looked like, where it lived, and why it is no longer alive.

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU USE TO MAKE YOUR MODEL FOSSIL — CLAY, PLASTER, OR SOMETHING ELSE?

2. WHAT INFORMATION WILL YOU INCLUDE ON THE EXHIBIT SIGN?

3. HOW WILL YOU SHOW WHAT THE ANIMAL LOOKED LIKE WHEN IT WAS ALIVE?

4. HOW WILL YOUR EXHIBIT EXPLAIN WHAT HAPPENED TO THE ANIMAL'S HABITAT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Paleontologists are scientists who study fossils to learn about plants and animals that lived millions of years ago. They dig up fossils, study them, and teach us about ancient life. They earn $50,000–$95,000/year.